Richardson method

find $x \in$ in $\mathbb{R}^n$ s.t.

$$Ax = b$$

where $A \in \mathbb{R}^{n \times n}$ S.P.D.

$$x^{(k+1)} = x^{(k)} + \alpha r^{(k)} \qquad \alpha \in \mathbb{R}$$

$$r^{(k)} = b - Ax^{(k)}$$

stationary Richardson method

$$\alpha < \frac{2}{\lambda_{max}(A)} \qquad \alpha_{opt} = \frac{2}{\lambda_{max}(A) + \lambda_{min}(A)}$$

Dynamic Richardson method

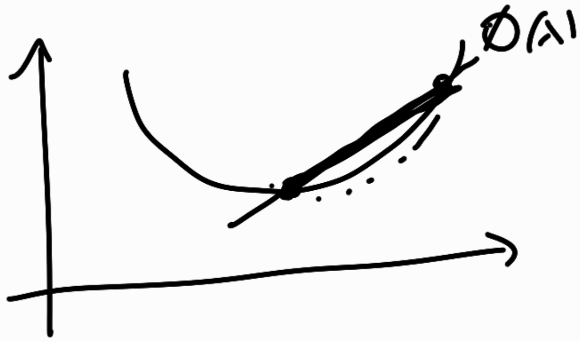
$$x^{(k+1)} = x^{(k)} + \underline{\underline{\alpha_k}} r^{(k)}$$

"Energy" functional

$$\phi(x) = \frac{1}{2} x^T A x - x^T b$$

If A is S.P.D. $\Rightarrow \phi(x)$ is convex

$$\phi(\alpha x + (1-\alpha) y) \leq \alpha \phi(x) + (1-\alpha) \phi(y)$$



$\phi(x)$ has one and only one stationary point x^*

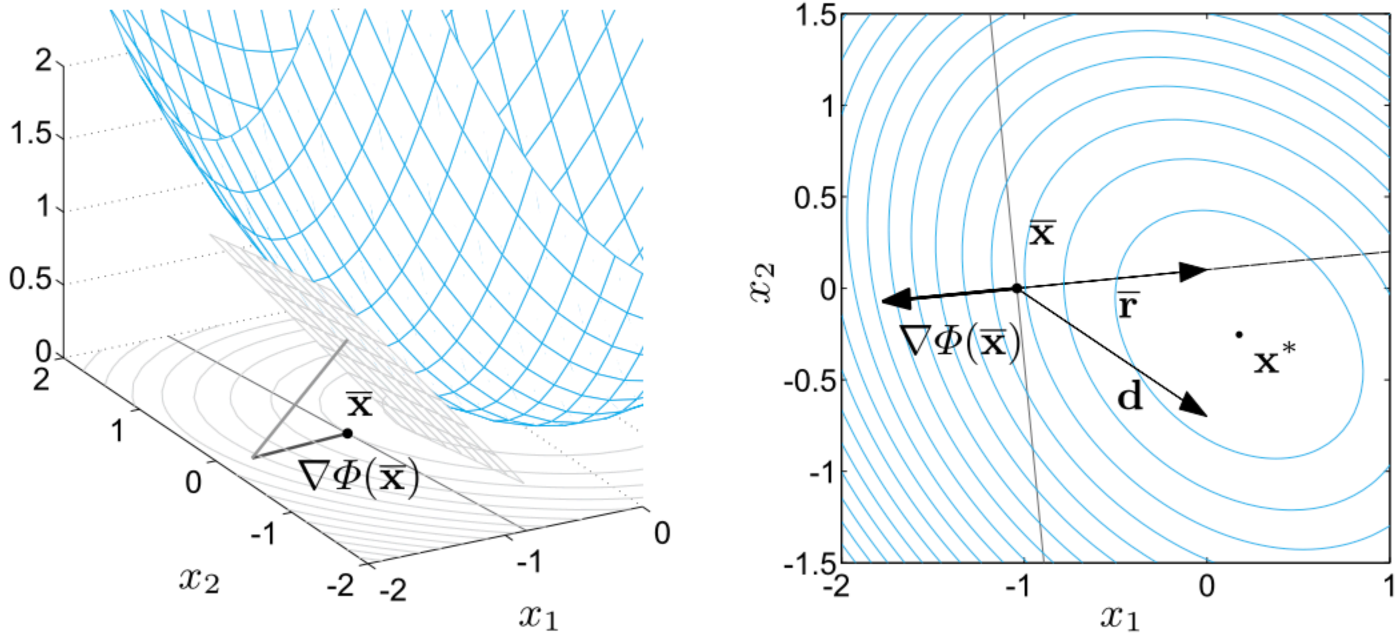
$$x^* = \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \ \phi(x)$$

$$\frac{\partial \phi}{\partial x} = \nabla_x \phi(x) = \nabla \phi(x) \in \mathbb{R}^n$$

$$\nabla \phi(x) = \begin{bmatrix} \frac{\partial \phi}{\partial x_1} \\ \frac{\partial \phi}{\partial x_2} \\ \vdots \end{bmatrix}$$

$$\nabla \phi(x) = Ax - b$$

find $x^* = \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \ \phi(x) \iff$ find x^* s.t. $Ax^* = b$



2
1.5
1
0.5
0
2
1
0
-1
-2
x_2
-2
-1
0
x_1
$\overline{\mathbf{x}}$
$\nabla\Phi(\overline{\mathbf{x}})$
1.5
1
0.5
0
-0.5
-1
-1.5
x_2
-2
-1
0
1
x_1
$\overline{\mathbf{x}}$
$\overline{\mathbf{r}}$
$\nabla\Phi(\overline{\mathbf{x}})$
$\mathbf{d}$
$\mathbf{x}^*$

Gradient method

$$x^{(k+1)} = x^{(k)} - \alpha_k \nabla \phi(x^{(k)})$$

$$\alpha_k = \underset{\alpha \in \mathbb{R}}{\operatorname{argmin}} \; \phi(x^{(k)} + \alpha \underbrace{d^{(k)}}_{-\nabla \phi(x^{(k)})})$$

$$\alpha_k = \frac{(d^{(k)})^T r^{(k)}}{(d^{(k)})^T A d^{(k)}}$$

Implementation of the Gradient method

given $x^{(0)}$, $r^{(0)} = b - Ax^{(0)}$, for $k = 0, 1, \dots$

$$\alpha_k = \frac{(r^{(k)})^T r^{(k)}}{(r^{(k)})^T A r^{(k)}}$$

$$x^{(k+1)} = x^{(k)} + \alpha_k r^{(k)}$$

$$r^{(k+1)} = r^{(k)} - \alpha_k A r^{(k)}$$

Properties of the Gradient method

- If A is S.P.D. the G.M. always converges ($\forall x^{(0)} \in \mathbb{R}^n$)

$$\| e^{(k)} \|_A \leq \left(\frac{K(A) - 1}{K(A) + 1} \right)^k \| e^{(0)} \|_A$$

$K(A)$ = 2-norm condition number

Directions of descent

$$x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)} = x^{(k-1)} + \alpha_{k-1} d^{(k-1)} + \\ + \alpha_k d^k$$

$$= x^{(0)} + \sum_{j=0}^{k} \alpha_j d^{(j)}$$

for the gradient method

$$(d^{(k)})^T r^{(k)} = 0 \ \forall$$

If the increment directions $d^{(j)}$ where all linearly independent we could express any vector in $\mathbb{R}^n$ as their linear combination

$$x^{(n)} = x^{(0)} + \sum_{j=0}^{n-1} \underline{\alpha_j} d^{(j)}$$

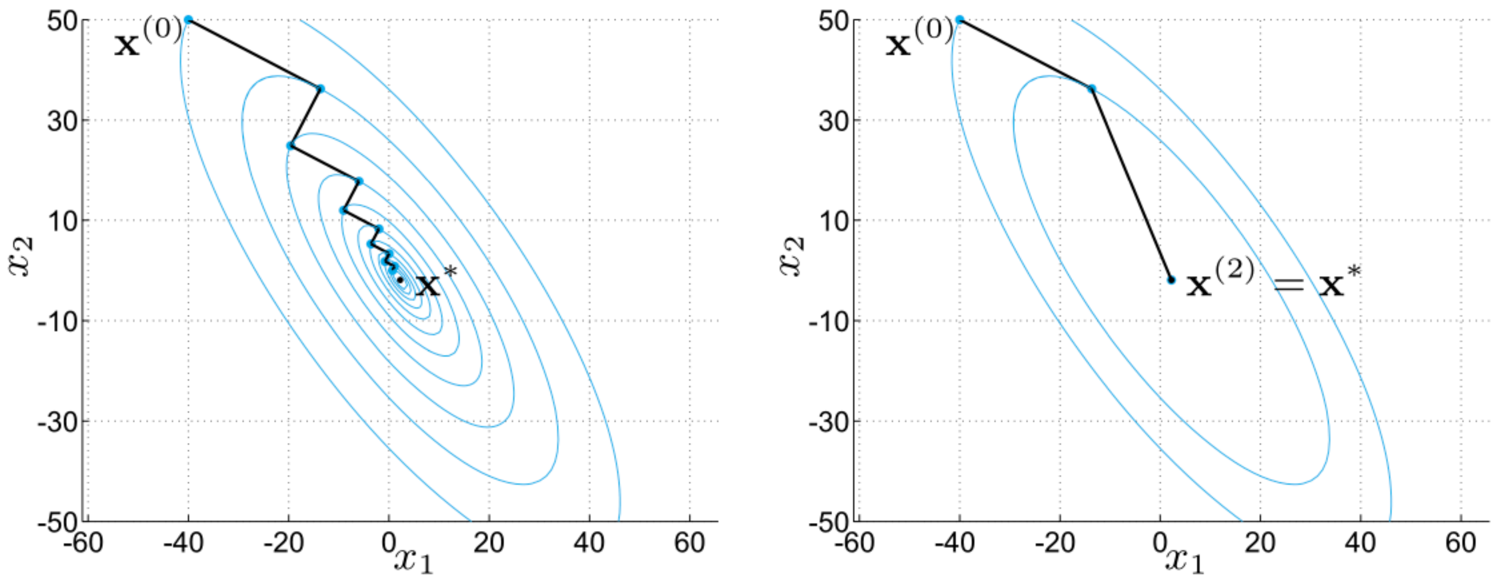
Conjugate gradient method

$$d^{(k+1)} = r^{(k+1)} - \beta_k d^{(k)} \qquad (*)$$

$$\beta_k = \frac{(Ad^{(k)})^T r^{(k+1)}}{(Ad^{(k)})^T d^{(k)}}$$

By constructing the sequence of descent directions according to $(*)$

- All $d^{(j)}$ are linearly independent
- $(r^{(k+1)})^T d^{(j)} = 0 \qquad j = 0, 1, \dots, k$
- After n iterations $r^{(n)}$ is orthogonal to all vectors in $\mathbb{R}^n \Rightarrow r^{(n)} = 0$



$\mathbf{x}^{(0)}$
$\mathbf{x}^*$
x_1
x_2
50
30
10
-10
-30
-50
-60
-40
-20
0
20
40
60
$\mathbf{x}^{(0)}$
$\mathbf{x}^{(2)} = \mathbf{x}^*$
x_1
x_2

For the C.G.

$$\|x\|_A = x^T A x$$

$$\|e^{(k)}\|_A \leq \frac{2\,c^k}{1+c^{2k}} \|e^{(0)}\|$$

$$c = \frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1}$$

$$\|e^{(n)}\| = 0 \quad \text{in exact arithmetics}$$

Implementation of the C.G. is as follows

$$\alpha_k = \frac{(d^{(k)})^T r^{(k)}}{(Ad^{(k)})^T d^{(k)}}$$

$$x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}$$

$$r^{(k+1)} = r^{(k)} - \alpha_k A d^{(k)}$$

$$\beta_k = \frac{(Ad^{(k)})^T r^{(k+1)}}{(Ad^{(k)})^T d^{(k)}}$$

$$d^{(k+1)} = d^{(k)} - \beta_k d^{(k)}$$

As the G.M. and C.G. have convergence rates that depend on the condition number of the matrix A $\Rightarrow$ to speed up convergence we can apply both methods to the "preconditioned" system

$$\underbrace{P^{-1}}_{\text{preconditioner}} A x = P^{-1} b$$

preconditioner

$P^{-1}A$ <u>should be S.P.D.</u> and well conditioned of course one does not compute

$P^{-1}A$ but instead in the

implementation of the method

we replace $r^{(k+1)}$ by the

pre-conditioned residual $z^{(k+1)}$

s.t.

$$\underline{Pz^{(k+1)}} = \underline{\underline{r^{(k+1)}}}$$